

Unit 3:

Cylindrical Coordinate System

What We'll Cover

- **Get to Know Cylindrical Coordinates:** Explore what the cylindrical coordinate system is, why it's important, and when it's useful.
- **Working with Position Vectors:** Learn how to express the position of a point using (ρ, ϕ, z) and relate this to rectangular coordinates.
- **Coordinate Surfaces and Unit Vectors:** Understand surfaces of constant ρ, ϕ, z and how the unit vectors $\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z$ behave.
- **Differential Elements:** Discover line, surface, and volume elements in cylindrical coordinates.
- **Applications in Engineering and Physics:** See how cylindrical coordinates simplify problems with cylindrical symmetry, like fields around wires or heat flow in pipes.

What You'll Be Able to Do by the End of This Presentation

- Describe what the cylindrical coordinate system is and when it should be used.
- Pinpoint locations in space using cylindrical coordinates.
- Convert between rectangular (x, y, z) and cylindrical (ρ, ϕ, z) coordinates.
- Visualize and interpret cylindrical coordinate surfaces.
- Apply cylindrical coordinates to solve practical problems in physics and engineering.

The Need for Cylindrical Coordinates

Many physical systems are best described not with straight lines (as in Cartesian coordinates), but with **rotational or radial symmetry**.

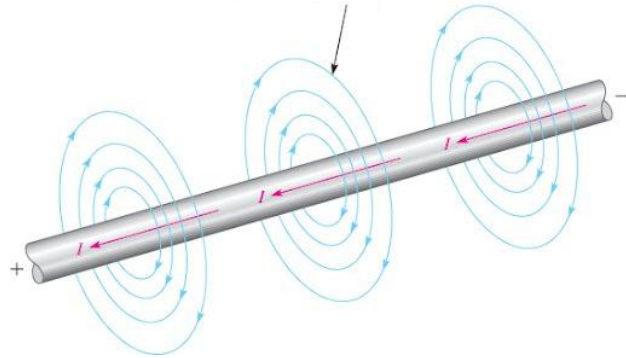
Examples:

- Magnetic fields circling a straight current-carrying conductor
- Temperature distribution in a cylindrical rod
- Fluid flow inside pipes

(see Figure 1)

Cylindrical coordinates provide a **natural framework** to model these problems.

Magnetic lines of force along conductor



(i) Magnetic fields circling a straight current-carrying conductor

(ii) Heat conduction across cylindrical walls.

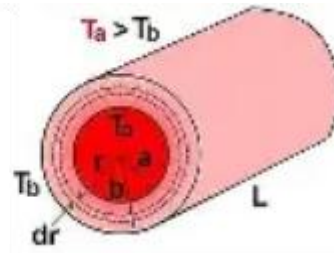
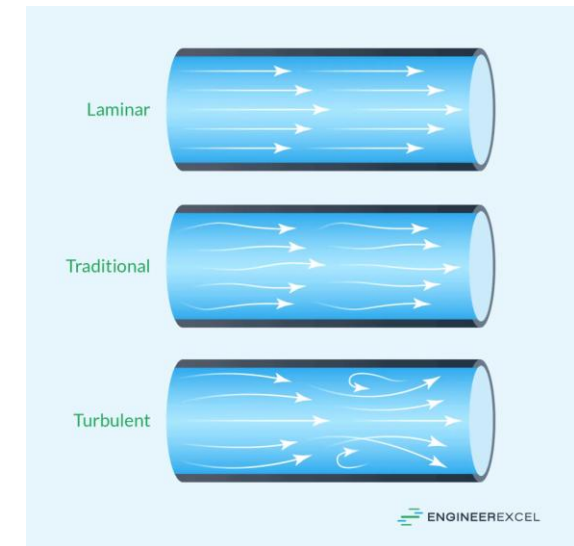


Figure 1



(iii) Different types of fluid flow in pipes.

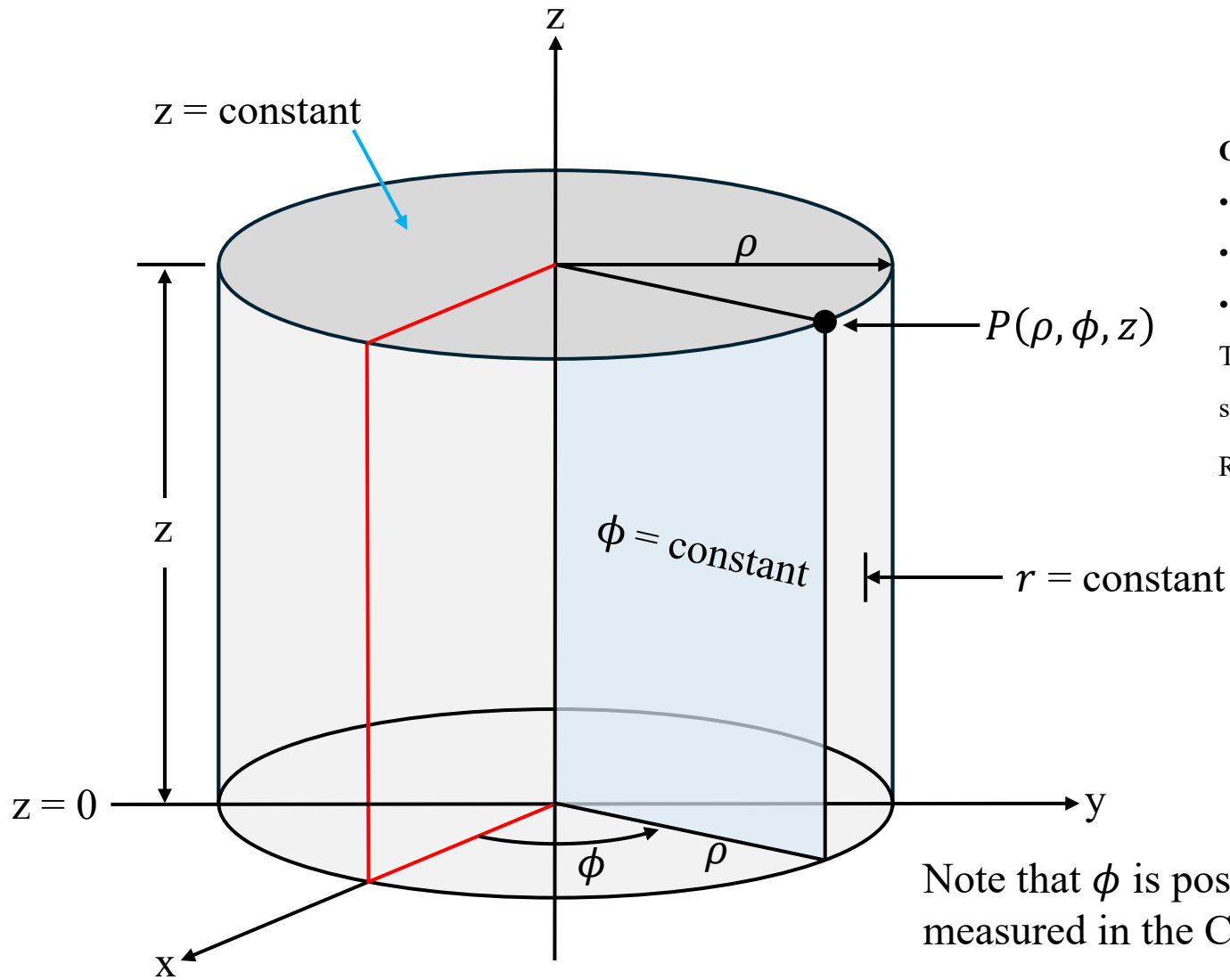
The Cylindrical Coordinate System

In the cylindrical coordinate system, the position of any point P in space is specified by three quantities that arise from the intersection of **three mutually perpendicular coordinate surfaces** (see Figure 2):

- ρ : the **radial distance** from the z-axis to the projection of the point in the xy -plane.
- ϕ : the **azimuthal angle**, measured in the xy -plane from the positive x-axis to the projection of the point.
- z : the **vertical coordinate**, the same as in the Cartesian (rectangular) system.

Thus, the coordinates of the point P are expressed in the order:

$$P(\rho, \phi, z)$$



Coordinate Surfaces

- $\rho = \text{constant}$ → a circular cylinder
- $\phi = \text{constant}$ → a vertical half-plane through the z -axis
- $z = \text{constant}$ → a horizontal plane

Together, these surfaces form a natural grid adapted to cylindrical shapes.

Ranges: $0 \leq \rho \leq \infty$
 $0 \leq \phi \leq 2\pi$
 $-\infty \leq z \leq \infty$

Note that ϕ is positive when measured in the CCW direction.

Figure 2: The cylindrical coordinate system.

A Note on Notation

The notation for cylindrical coordinates in the literature is **not uniform**.

The **ISO standard 31-11** recommends (ρ, ϕ, z) , where:

- ρ = radial distance
- ϕ = azimuthal angle
- z = height

However, in much of the engineering and physics literature, it is also common to see:

- r instead of ρ (radial coordinate)
- θ instead of ϕ (azimuthal angle)

Because of this inconsistency, one must read carefully to avoid confusion.

In this course, for clarity and consistency with common engineering practice, we will adopt the notation prescribed by **ISO Standard 31-11**.

Figure 3 provides a visual comparison of the two conventions, showing the **ISO standard notation** (ρ, ϕ, z) used in this course side by side with the **alternative notation** (r, θ, z) often found in other references.

Figure 3

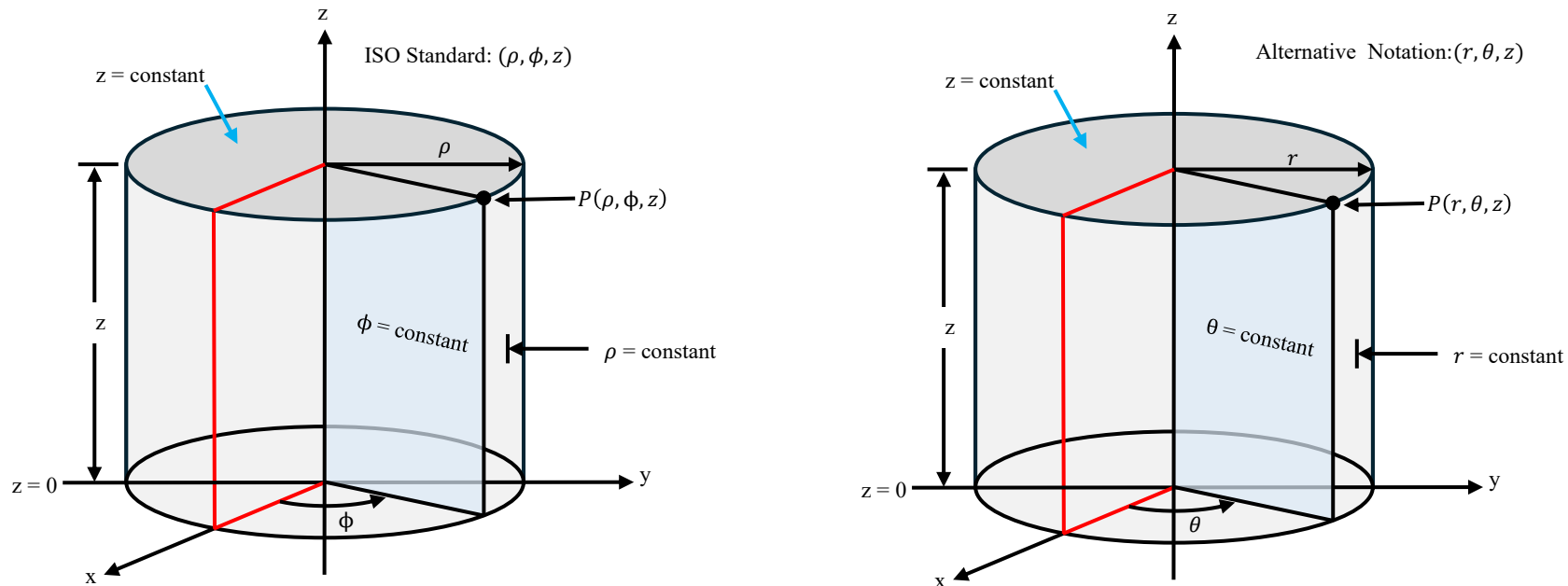


Figure 4 shows a visualization of point P ($3, 45^\circ, 8$) in the cylindrical coordinate system.

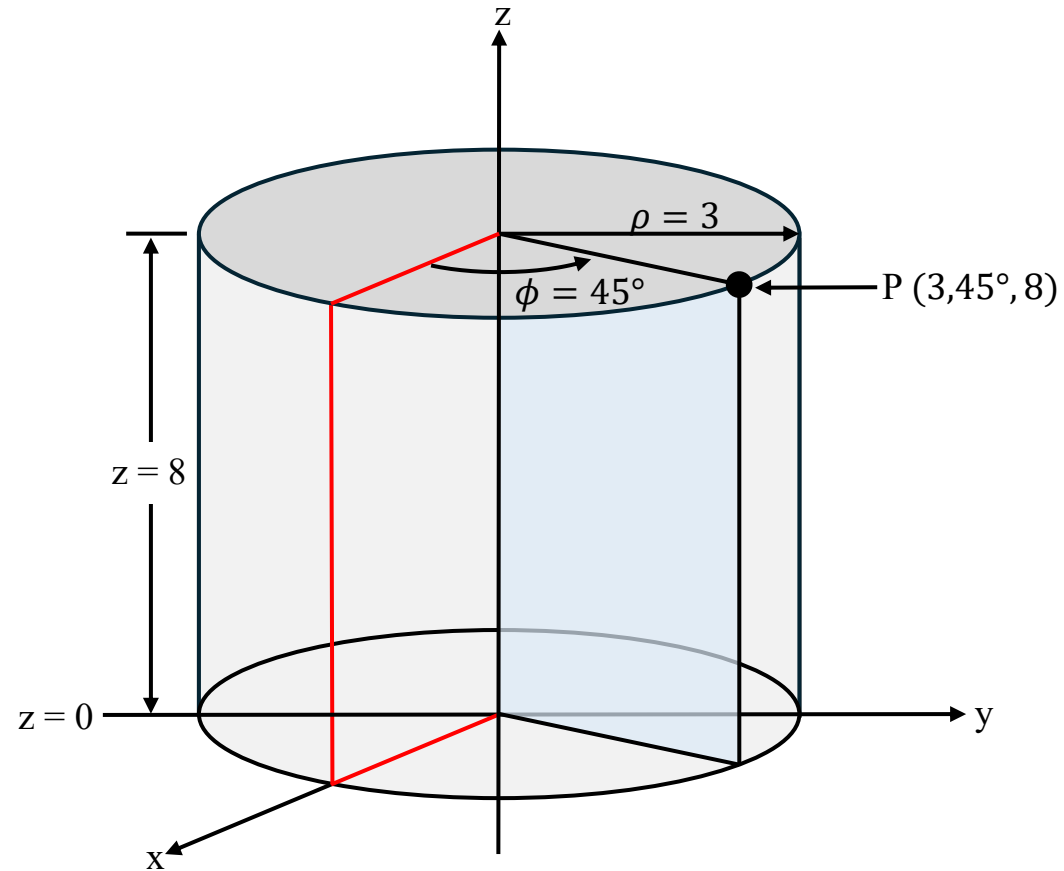


Figure 4

Coordinate Transformations: Cartesian $\leftrightarrow$ Cylindrical

To convert the coordinates of a point $P(x, y, z)$ between the **Cartesian** (x, y, z) system and the **cylindrical** (ρ, ϕ, z) system, we use the following transformation equations:

From Cartesian to Cylindrical	From Cylindrical to Cartesian
$\rho = \sqrt{x^2 + y^2}$	$x = \rho \cos \phi$
$\phi = \tan^{-1}(y/x)$	$y = \rho \sin \phi$
$z = z$	$z = z$

(see Figure 5)

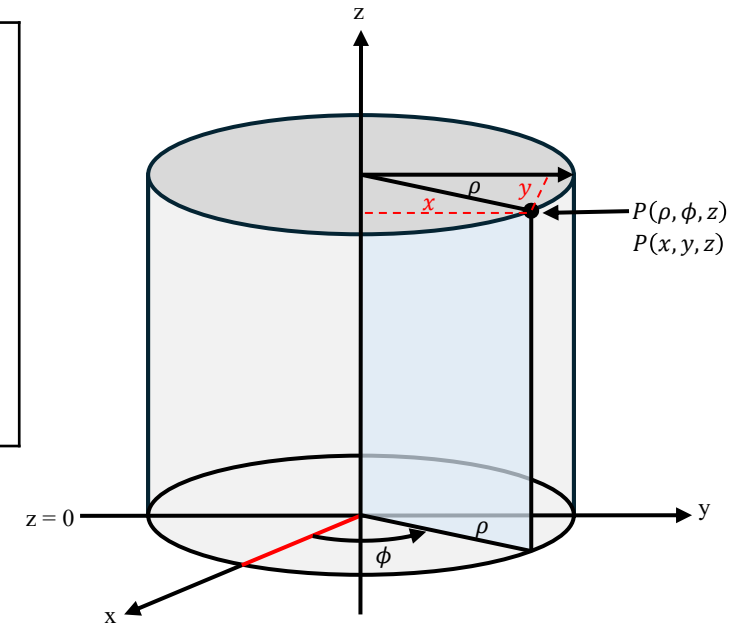


Figure 5

Example: Cartesian $\rightarrow$ Cylindrical Conversion

Problem:

Convert the point

$$P(x, y, z) = (3, 4, 5)$$

from **Cartesian coordinates** to **cylindrical coordinates** (ρ, ϕ, z) .

(See Figure 5)

Solution

Step 1. Compute the radial distance r (see Figure 6)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = 5$$

Step 2. Compute the azimuthal angle θ

$$\phi = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{4}{3} \right) \approx 53.13^\circ$$

Step 3. Keep the z-coordinate

$$z = 5$$

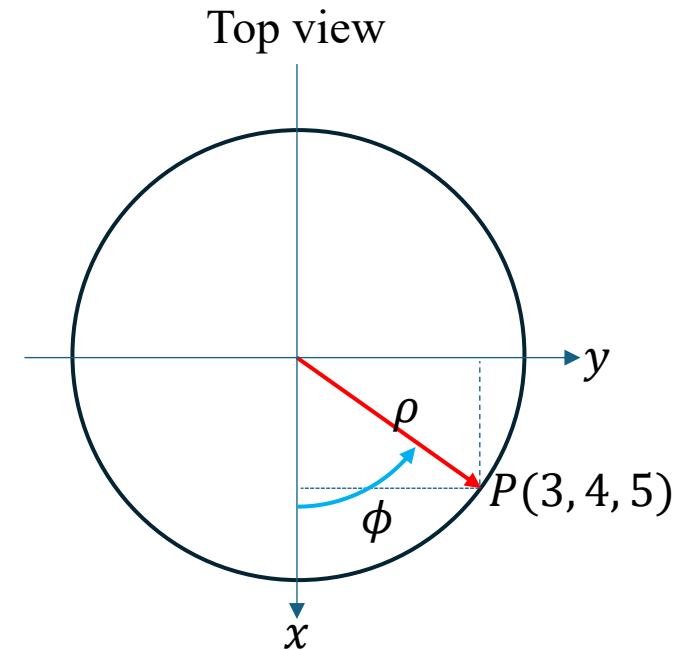


Figure 6

Final Answer

The point in cylindrical coordinates is:

$$P(\rho, \phi, z) = (5, 53.13^\circ, 5)$$

(see Figure 7)

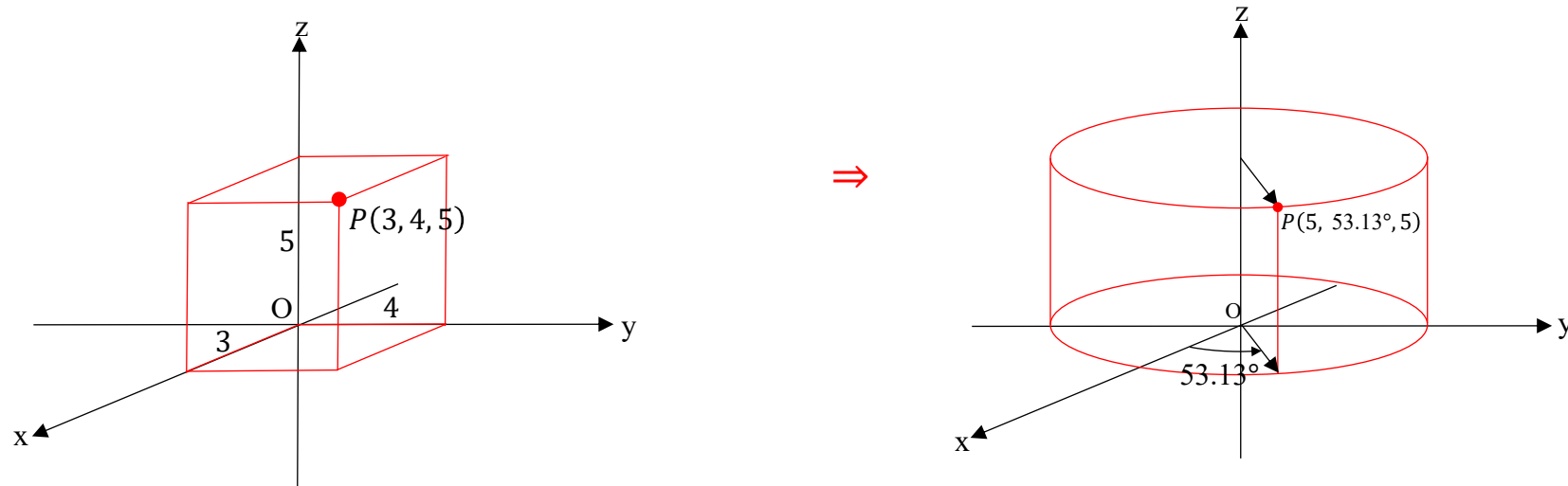


Figure 7

Example 2: Cartesian $\rightarrow$ Cylindrical Conversion

Problem:

Convert the point

$$P(x, y, z) = (-3, 4, 2)$$

from **Cartesian coordinates** to **cylindrical coordinates** (ρ, ϕ, z) .

Step 1. Compute the radial distance r (See Figure 8)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + 4^2} = 5$$

Step 2. Compute the azimuthal angle θ

$$\phi = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{4}{-3} \right) = -53.13^\circ$$

But since the point lies in **Quadrant II** ($x < 0, y > 0$),
we must add 180° (or π radians):

$$\phi = -53.13^\circ + 180^\circ = 126.87^\circ$$

3. Keep the z-coordinate

$$z = 2$$

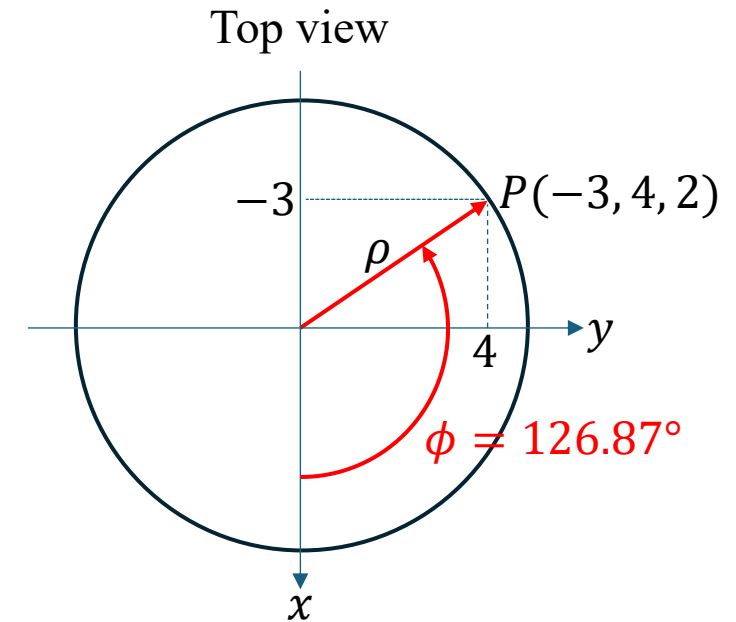


Figure 8

Final Answer

The point in cylindrical coordinates is:

$$P(\rho, \phi, z) = (5, 126.87^\circ, 2)$$

(see Figure 9)

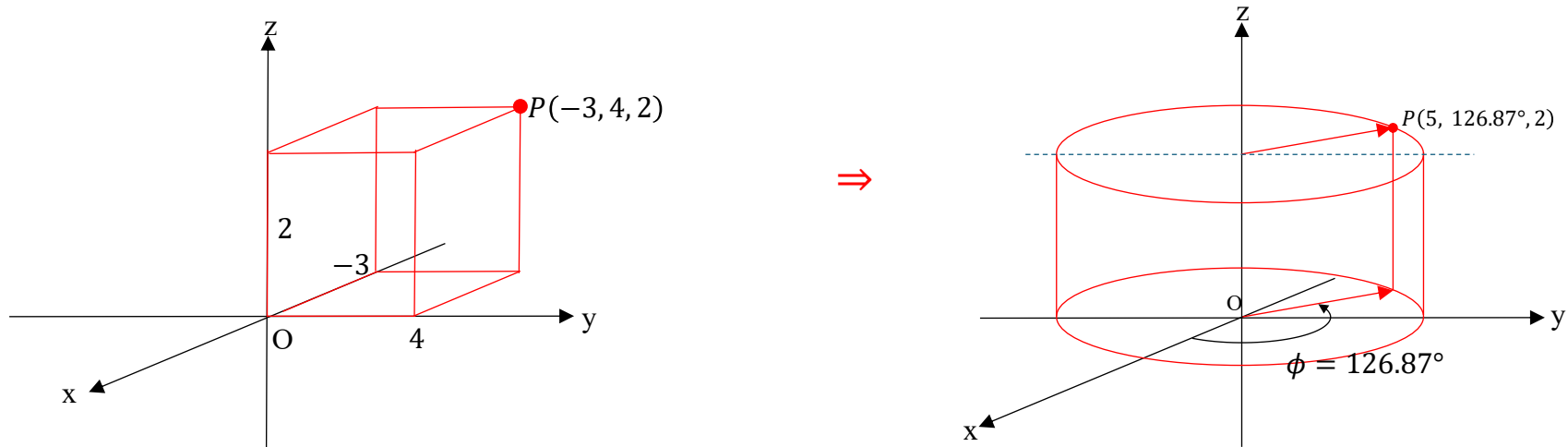


Figure 9

Example: Polar $\rightarrow$ Cartesian Conversion

Problem:

Convert the point in **polar coordinates**

$$P(\rho, \phi, z) = (4, \frac{5\pi}{6}, 3)$$

to **Cartesian coordinates** (x, y, z) .

Step 1. Recall the transformation equations

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

Step 2. Substitute the values

$$x = 4 \cos \left(\frac{5\pi}{6} \right) = 4 \left(-\frac{\sqrt{3}}{2} \right) = -2\sqrt{3}$$

$$y = 4 \sin \left(\frac{5\pi}{6} \right) = 4 \left(\frac{1}{2} \right) = 2$$

$$z = 3$$

(see Figure 10)

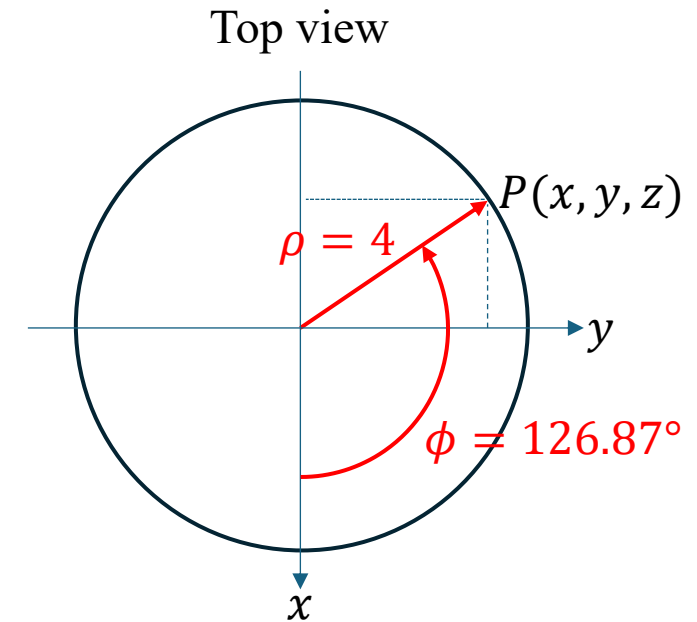


Figure 10

Final Answer

The Cartesian coordinates are:

$$P(x, y) = (-2\sqrt{3}, 2, 3)$$

(see Figure 11)

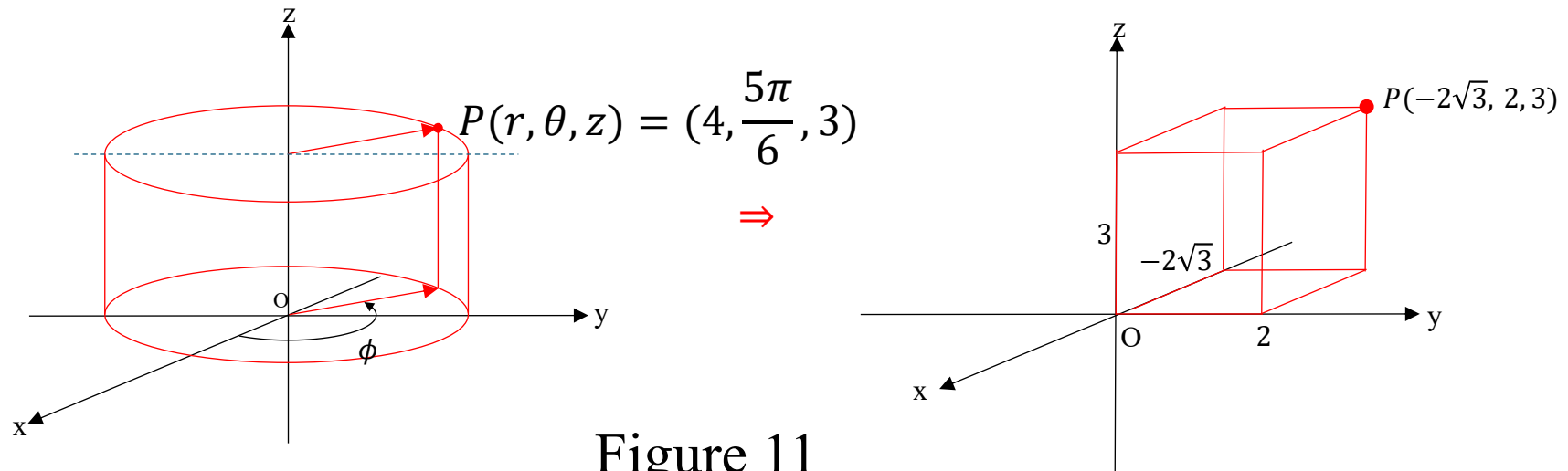


Figure 11

Position Vectors in the Cylindrical Coordinate System

No matter what coordinate system we use, a **position vector** must always be expressed in terms of the **Cartesian base vectors** $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$.

Why only Cartesian base vectors?

- Cartesian base vectors have **fixed directions** in space.
- In contrast, cylindrical (or spherical) base vectors change their direction depending on the position of the point.
- Thus, for consistency, the position vector is always built using $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$.

Position Vector in Cylindrical Coordinates Using Cartesian Base Vectors

For a point $P(\rho, \phi, z)$ in cylindrical coordinates (see Figure 12), the position vector is:

$$\mathbf{R} = \overrightarrow{OP} = \rho \cos \phi \hat{\mathbf{i}} + \rho \sin \phi \hat{\mathbf{j}} + z \hat{\mathbf{k}}$$

In column vector form, we write:

$$\mathbf{R} = \begin{bmatrix} \rho \cos \phi \\ \rho \sin \phi \\ z \end{bmatrix}$$

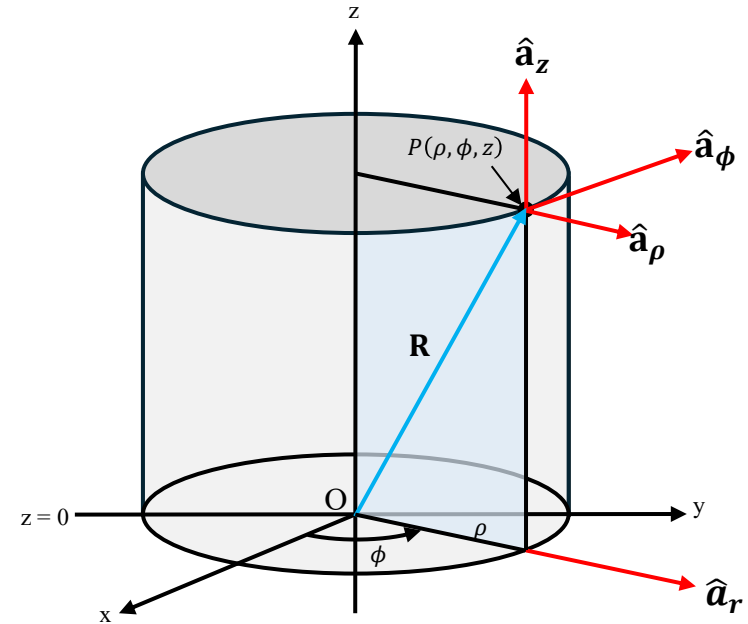


Figure 12

Position Vectors in Terms of Cylindrical Base Vectors

In cylindrical coordinates, we use the **moving base vectors**:

$$\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z$$

- $\hat{\mathbf{a}}_\rho$: points outward in the radial direction.
- $\hat{\mathbf{a}}_\phi$: tangent to the circle in the xy -plane, perpendicular to $\hat{\boldsymbol{\rho}}$.
- $\hat{\mathbf{a}}_z$: same as Cartesian $\hat{\mathbf{k}}$, pointing along the z -axis.

(see Figure 13)

The position vector of point P can be expressed as:

$$\mathbf{R} = \rho \hat{\mathbf{a}}_\rho + z \hat{\mathbf{a}}_z$$

where

$$|\mathbf{R}| = \sqrt{\rho^2 + z^2}, \quad \hat{\mathbf{a}}_\rho = \cos\phi \hat{\mathbf{i}} + \sin\phi \hat{\mathbf{j}}$$

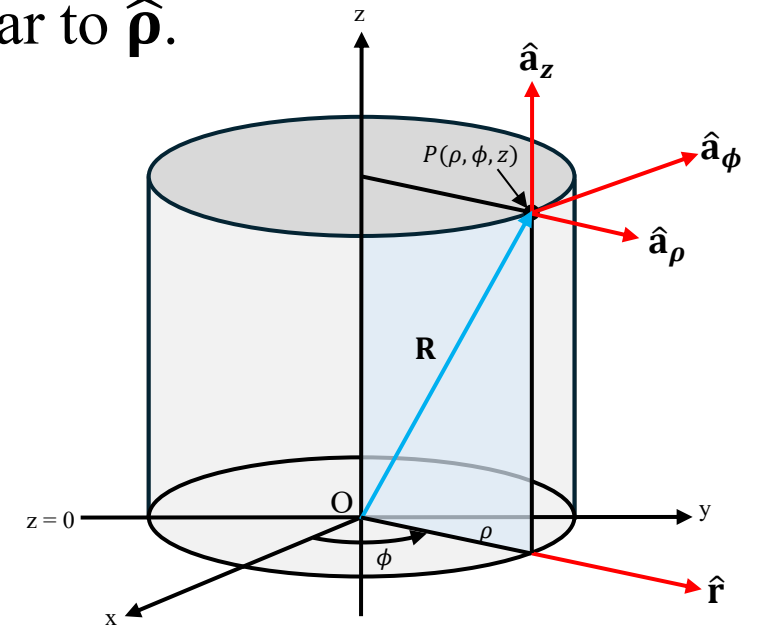


Figure 13

Example: Position Vector in Cylindrical Coordinates

Problem:

Find the **position vector** of a point P with cylindrical coordinates:

$$(\rho, \phi, z) = (4, 60^\circ, 5)$$

Express the vector in terms of:

- (a) the cylindrical base vectors $\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z$
- (b) Cartesian base vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}},$

Solution

(a) In cylindrical coordinates, the position vector is

$$\mathbf{R} = r\hat{\mathbf{a}}_\rho + z\hat{\mathbf{a}}_z \quad (\text{no } \hat{\mathbf{a}}_\theta \text{ component}).$$

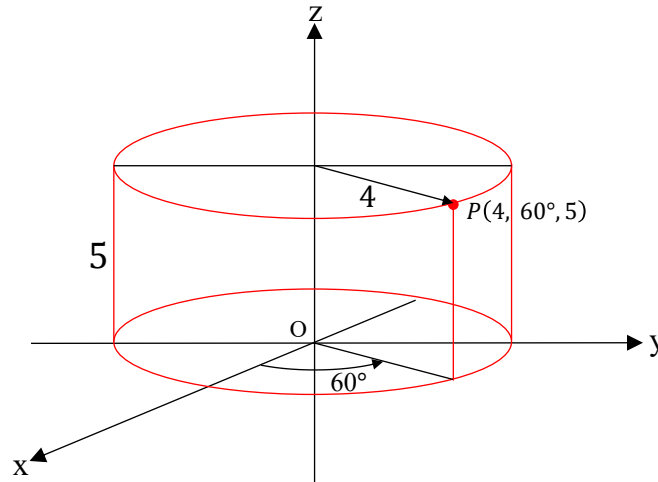
Given $(\rho, \phi, z) = (4, 60^\circ, 5)$,

$$\mathbf{R} = 4\hat{\mathbf{a}}_\rho + 0\hat{\mathbf{a}}_\phi + 5\hat{\mathbf{a}}_z$$

(ϕ affects the directions of $\hat{\mathbf{a}}_\rho$ and $\hat{\mathbf{a}}_\phi$, but not the components of $\mathbf{R}$ in the cylindrical basis.)

(see Figure 14)

Figure 14



(b) Recall the formula

$$\mathbf{R} = \rho \cos \phi \hat{\mathbf{i}} + \rho \sin \phi \hat{\mathbf{j}} + z \hat{\mathbf{k}}$$

Substitute the given values

$$\mathbf{R} = (4 \cos 60^\circ) \hat{\mathbf{i}} + (4 \sin 60^\circ) \hat{\mathbf{j}} + (5) \hat{\mathbf{k}}$$

Simplify

$$\cos 60^\circ = 0.5; \quad \sin 60^\circ = 0.866$$

$$\mathbf{R} = (4 \times 0.5) \hat{\mathbf{i}} + (4 \times 0.866) \hat{\mathbf{j}} + (5) \hat{\mathbf{k}}$$

$$\mathbf{R} = 2 \hat{\mathbf{i}} + 3.464 \hat{\mathbf{j}} + 5 \hat{\mathbf{k}}$$

Final Answer

The position vector in Cartesian base vectors is:

$$\mathbf{R} = 2 \hat{\mathbf{i}} + 3.464 \hat{\mathbf{j}} + 5 \hat{\mathbf{k}} \blacksquare$$

Distance Vector in Cylindrical Coordinates

Suppose we have two points in cylindrical coordinates (see Figure 15):

- $P(\rho_1, \phi_1, z_1)$, located by the position vector $\mathbf{R}_1$
- $Q(\rho_2, \phi_2, z_2)$, located by the position vector $\mathbf{R}_2$

The **distance vector** between these points is:

$$\overrightarrow{PR} = \mathbf{R}_2 - \mathbf{R}_1$$

The **distance** d separating the two points is the magnitude of this vector:

$$d = |\overrightarrow{PR}| = |\mathbf{R}_2 - \mathbf{R}_1|$$

Formula in Cylindrical Coordinates

$$d = |\overrightarrow{PQ}| = \sqrt{R_1^2 + R_2^2 - 2R_1R_2 \cos(\phi_2 - \phi_1) + (z_1 - z_2)^2}$$

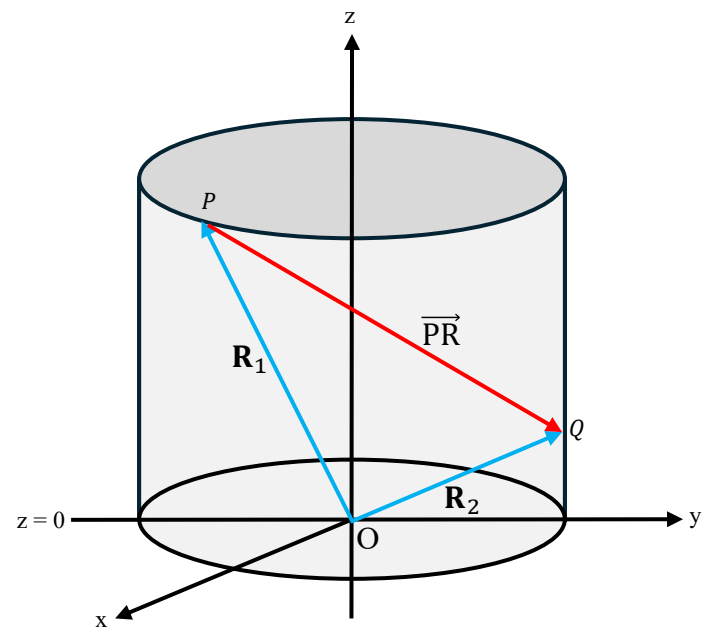


Figure 15

Alternative Method: Distance via Position Vectors

Another way to compute the distance between two points is to first express each point as a **position vector** in Cartesian components, then take the **magnitude of their difference**.

Suppose we have two points:

$$A(\rho_a, \phi_a, z_a), \quad B(\rho_b, \phi_b, z_b)$$

(See Figure 16.)

Their position vectors in **column vector form** are:

$$\mathbf{A} = \begin{bmatrix} \rho_a \cos \phi_a \\ \rho_a \sin \phi_a \\ z_a \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \rho_b \cos \phi_b \\ \rho_b \sin \phi_b \\ z_b \end{bmatrix}$$

Distance Formula

The distance d separating the two points is:

$$d = |\overrightarrow{AB}| = |\mathbf{A} - \mathbf{B}|$$

That is, the distance equals the **magnitude of the difference between the two position vectors**.

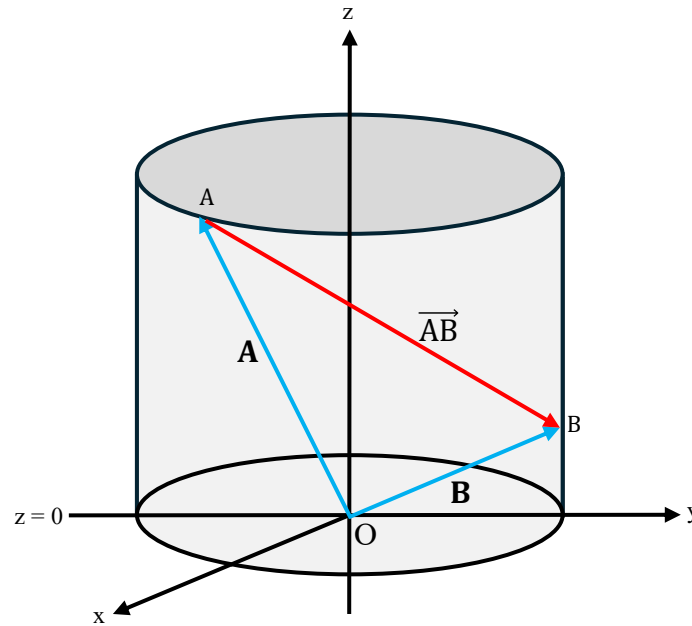


Figure 16

Problem

Find the distance between the two points, expressed in cylindrical coordinates.

$$A(\rho_a, \phi_a, z_a) = (3, 30^\circ, 2),$$

$$B(\rho_b, \phi_b, z_b) = (5, 120^\circ, 6)$$

Method 1 — Use the cylindrical distance formula

Recall the formula

$$d = \sqrt{\rho_a^2 + \rho_b^2 - 2\rho_a\rho_b \cos(\phi_b - \phi_a) + (z_b - z_a)^2}$$

1. Compute the angle difference:

$$\phi_b - \phi_a = 120^\circ - 30^\circ = 90^\circ$$

$$\cos(90^\circ) = 0.$$

2. Substitute numbers:

$$d = \sqrt{3^2 + 5^2 - 2 \cdot 3 \cdot 5 \cdot 0 + (6 - 2)^2} = \sqrt{50} \approx 7.0711.$$

Result:

$$d \approx 7.0711 \blacksquare$$

Method 2 — Convert to Cartesian position vectors, subtract, take magnitude

1. Convert A and B to Cartesian components using

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

- For A(3, 30°, 2):

$$x_a = 3 \cos 30^\circ = 3 \cdot \frac{\sqrt{3}}{2} \approx 2.598076,$$

$$y_a = 3 \sin 30^\circ = 3 \cdot \frac{1}{2} = 1.5, \quad z_a = 2.$$

- For B(5, 120°, 6):

$$x_b = 5 \cos 120^\circ = 5 \cdot \left(-\frac{1}{2}\right) = -2.5,$$

$$y_b = 5 \sin 120^\circ = 5 \cdot \frac{\sqrt{3}}{2} = 4.3330127, \quad z_b = 6.$$

2. Form the difference vector $\mathbf{A} - \mathbf{B}$:

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} x_a - x_b \\ y_a - y_b \\ z_a - z_b \end{bmatrix} = \begin{bmatrix} 2.598076 - (-2.5) \\ 1.5 - 4.330127 \\ 2 - 6 \end{bmatrix} = \begin{bmatrix} 5.098076 \\ -2.830127 \\ -4 \end{bmatrix}$$

3. Square the components and sum:

$$5.098076^2 \approx 25.9939$$

$$(-2.830127)^2 \approx 8.0061$$

$$(-4)^2 = 16$$

$$\text{Sum} \approx 25.9939 + 8.0061 + 16 = 50.0000.$$

4. Take the square root:

$$d = \sqrt{50} \approx 7.0711$$

Same result as Method 1 — good consistency check.

Exercise

Find the distance between the two points, expressed in cylindrical coordinates.

$$A(\rho_a, \phi_a, z_a) = (9.50 \text{ m}, 1.0053 \text{ rad}, 18.2 \text{ m}),$$

$$B(\rho_b, \phi_b, z_b) = (9.75 \text{ m}, 5.27788 \text{ rad}, 18.2 \text{ m})$$

Physical Vectors and Vector Fields in the Cylindrical Coordinate System

At a point $P(\rho, \phi, z)$ in the cylindrical coordinate system (see Figure 17), a physical vector $\mathbf{A}$ can be represented using three **mutually perpendicular unit vectors**:

$$\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z$$

- $\hat{\mathbf{a}}_\rho$: points outward in the radial direction (ρ).
- $\hat{\mathbf{a}}_\phi$: tangent to the circle in the azimuthal direction (ϕ).
- $\hat{\mathbf{a}}_z$: points vertically along the z-axis.

These three unit vectors form a **right-handed system**: if a right-hand screw is rotated from $\hat{\mathbf{a}}_\rho$ to $\hat{\mathbf{a}}_\phi$, it advances in the $\hat{\mathbf{a}}_z$ direction.

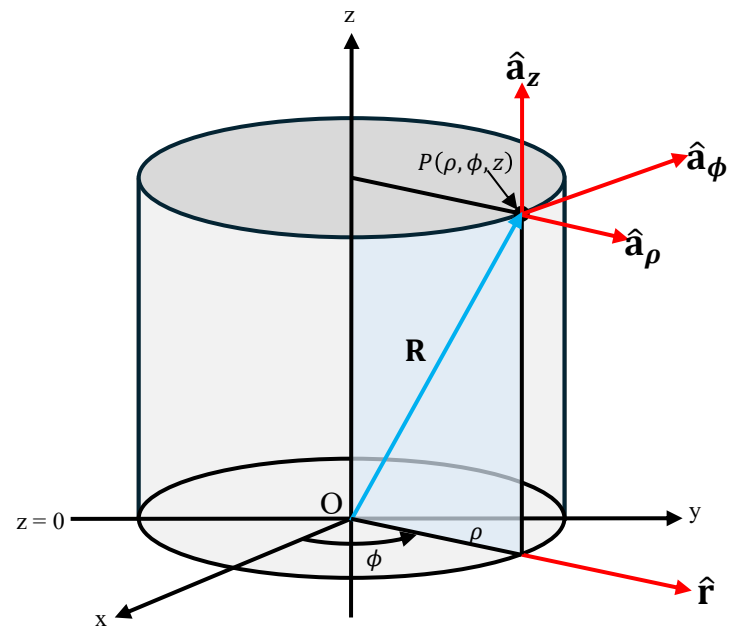


Figure 17

Cross-Product Relations

$$\hat{\mathbf{a}}_\rho \times \hat{\mathbf{a}}_\phi = \hat{\mathbf{a}}_z,$$

$$\hat{\mathbf{a}}_\phi \times \hat{\mathbf{a}}_z = \hat{\mathbf{a}}_\rho,$$

$$\hat{\mathbf{a}}_z \times \hat{\mathbf{a}}_\rho = \hat{\mathbf{a}}_\phi$$

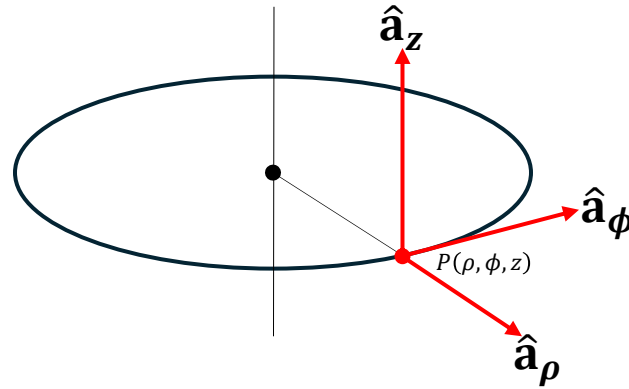


Figure 18

Vector Representation

Using these unit vectors, a general vector $\mathbf{A}$ at point P (see Figure 18) is expressed as:

$$\mathbf{A} = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

where A_ρ , A_ϕ , A_z are the scalar components of $\mathbf{A}$ in the cylindrical directions.

1. Notes

Some literature adopts an alternative notation for the rectangular unit vectors, using $\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_y$, $\hat{\mathbf{a}}_z$ in place of the more common $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ symbols. Thus, instead of writing a physical vector $\mathbf{A}$ as

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}},$$

it may alternatively be expressed as

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z.$$

Both notations are equivalent and represent the same vector components along the $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ axes, respectively. The choice of notation typically depends on the convention used in a particular textbook or discipline.

Notes (continued)

2. Just like $(\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$ in Cartesian coordinates, the cylindrical base vectors form a complete, mutually perpendicular set.
3. The main difference is that $\hat{\mathbf{a}}_\rho$, $\hat{\mathbf{a}}_\phi$, and $\hat{\mathbf{a}}_z$ **depend on position** (they rotate as ϕ changes).

Representation of Physical Vectors in the Cylindrical Coordinate System

When solving engineering problems, it is often advantageous to describe physical quantities such as electric fields, magnetic flux density, or velocity in a coordinate system that reflects the geometry of the physical situation. Many systems encountered in electrical and mechanical engineering—such as coaxial cables, rotating machinery, solenoids, and circular pipes—possess **axial or rotational symmetry**. For such configurations, the **cylindrical coordinate system** provides a convenient mathematical framework.

In the cylindrical coordinate system, a point in space is located by the ordered triple (ρ, ϕ, z) :

- The coordinate ρ specifies the perpendicular distance from the point to the z -axis.
- The coordinate ϕ gives the angular displacement of the point's projection on the x – y plane, measured from the positive x -axis.
- The coordinate z retains the same meaning as in Cartesian coordinates, representing the height of the point above (or below) the reference plane.

The geometry of this coordinate system is defined by three mutually perpendicular **unit vectors**:

$$\mathbf{a}_\rho, \mathbf{a}_\phi, \mathbf{a}_z$$

Each points in the direction of increasing ρ , ϕ , and z , respectively. Unlike the fixed unit vectors $\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$ in rectangular coordinates, the vectors $\mathbf{a}_\rho$ and $\mathbf{a}_\phi$ **rotate** as the point moves around the z -axis. Their orientation therefore depends on the angular position ϕ .

Cylindrical-to Rectangular Coordinate Transformation

Suppose we are given a physical vector $\mathbf{A}(x, y, z)$ —for example, the repulsive force experienced by a charged particle at point P due to another charge at point Q (see Figure 20).

In the **rectangular (Cartesian)** coordinate system, the vector can be expressed as:

$$\mathbf{A}(x, y, z) = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z.$$

Alternatively, in the **cylindrical** coordinate system, the same vector is expressed as:

$$\mathbf{A}(\rho, \phi, z) = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

This transformation allows us to choose the most convenient representation depending on the **symmetry of the physical problem**.

Rectangular-to-Cylindrical Coordinate Transformation

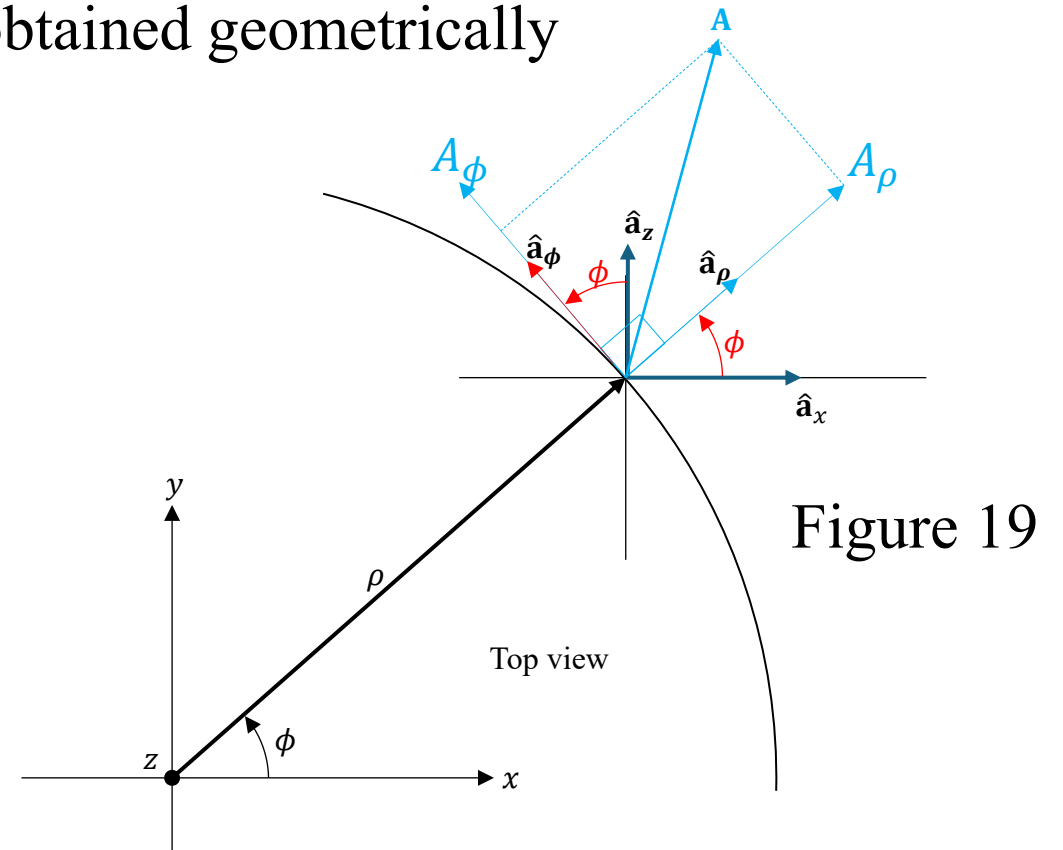
Relationship Between Unit Vectors

The relationships between the rectangular unit vectors $(\hat{\mathbf{a}}_x, \hat{\mathbf{a}}_y, \hat{\mathbf{a}}_z)$ and the cylindrical unit vectors $(\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z)$ can be obtained geometrically from Figure 19. They are given by:

$$\hat{\mathbf{a}}_\rho = \cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_\phi = -\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_z = \hat{\mathbf{a}}_z$$



The vector $\mathbf{A}(\rho, \phi, z)$ expressed in cylindrical coordinates can therefore be written as:

$$\mathbf{A}(\rho, \phi, z) = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

Substituting for $\hat{\mathbf{a}}_\rho, \hat{\mathbf{a}}_\phi, \hat{\mathbf{a}}_z$ in terms of rectangular unit vectors gives:

$$\mathbf{A}(\rho, \phi, z) = A_\rho (\cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y) + A_\phi (-\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y) + A_z \hat{\mathbf{a}}_z$$

Simplifying,

$$\mathbf{A}(\rho, \phi, z) = (A_\rho \cos \phi - A_\phi \sin \phi) \hat{\mathbf{a}}_x + (A_\rho \sin \phi + A_\phi \cos \phi) \hat{\mathbf{a}}_y + A_z \hat{\mathbf{a}}_z$$

Matrix Representation

In matrix form, the transformation between **rectangular** and **cylindrical** components is expressed as:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

or equivalently,

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Compact Vector Form

Symbolically, this relationship can be written as:

$$\mathbf{A}(\rho, \phi, z) = [T] \mathbf{A}(x, y, z)$$

where

- $\mathbf{A}(\rho, \phi, z)$ = vector $\mathbf{A}$ expressed in **cylindrical components**,
- $\mathbf{A}(x, y, z)$ = vector $\mathbf{A}$ expressed in **rectangular components**, and
- $[T]$ is the **transformation matrix** relating the two coordinate systems.

This transformation provides a systematic way to express any physical vector—such as an **electric field**, **magnetic field**, or **force vector**—in the coordinate system that best matches the **geometry and symmetry** of the problem.

Example: Converting Rectangular Components to Cylindrical Vector Components

Problem

Suppose a vector field represents the **magnetic flux density** at a certain point in space, given in **rectangular coordinates** at the spatial location $\phi = 45^\circ$ as

$$\mathbf{B} = 3.5355 \hat{\mathbf{a}}_x - 0.7071 \hat{\mathbf{a}}_y + 4.0000 \hat{\mathbf{a}}_z \text{ (mT)}$$

Determine the components of this vector in the **cylindrical coordinate system**.

(see Figure 20)

Solution

Step 1: Recall the transformation relationships

The cylindrical components (B_ρ, B_ϕ, B_z) are related to the rectangular components (B_x, B_y, B_z) by

$$B_\rho = B_x \cos \phi + B_y \sin \phi$$

$$B_\phi = -B_x \sin \phi + B_y \cos \phi$$

$$B_z = B_z$$

These relations may also be expressed in matrix form as

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix}$$

Given

$$B_x = 3.5355, \quad B_y = -0.7071, \quad B_z = 4.0000, \quad \phi = 45^\circ$$

and

$$\cos 45^\circ = \sin 45^\circ = 0.7071$$

Substitute these into the equations:

$$B_r = (3.5355)(0.7071) + (-0.7071)(0.7071) = 2.000$$

$$B_\phi = -(3.5355)(0.7071) + (-0.7071)(0.7071) = -3.000$$

$$B_z = 4.0000$$

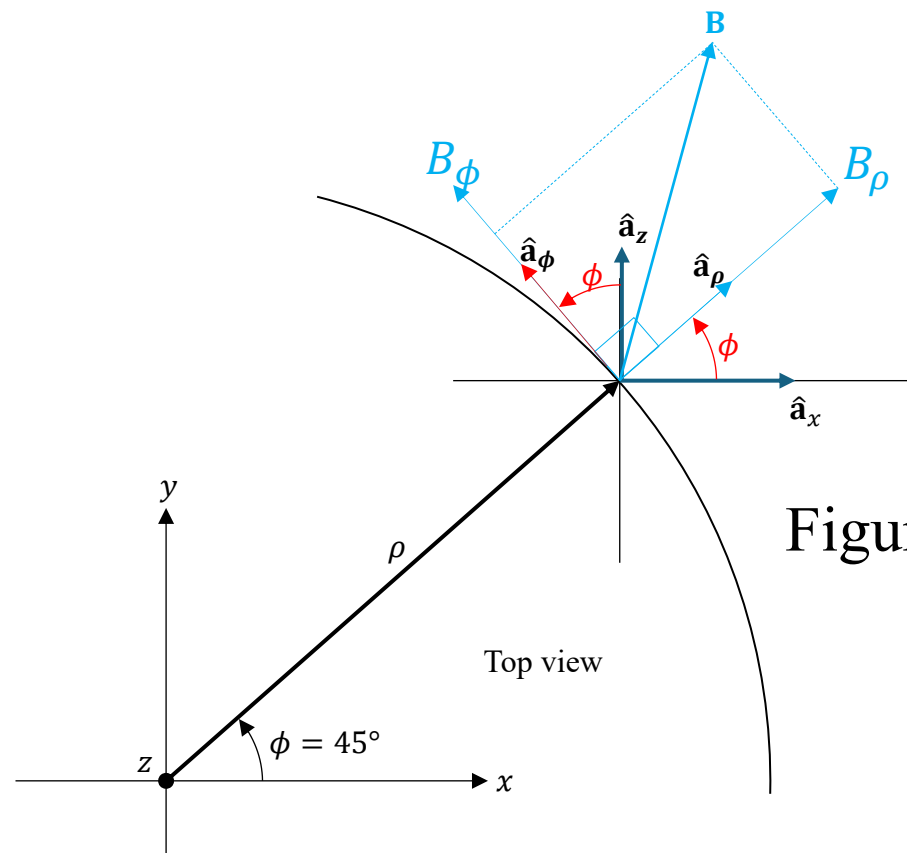


Figure 20

Step 3: Compute the results

$$B_\rho = 2.0000$$

$$B_\phi = -3.0000$$

$$B_z = 4.0000$$

Step 4: Express the final vector

$$\mathbf{B} = 2.00 \hat{\mathbf{a}}_\rho - 3.00 \hat{\mathbf{a}}_\phi + 4.00 \hat{\mathbf{a}}_z \text{ (mT)} \blacksquare$$

Step 5. Matrix Representation

Alternatively, this conversion can be expressed compactly in **matrix form**:

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix}$$

At $\phi = 45^\circ$:

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} 0.7071 & 0.7071 & 0 \\ -0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3.5355 \\ -7.071 \\ 4.000 \end{bmatrix} = \begin{bmatrix} 2.000 \\ -3.000 \\ 4.000 \end{bmatrix} \text{ (mT)}$$

which agrees with the direct computation.

Cylindrical-to-Rectangular Coordinate Transformation

Just as we can express a rectangular vector in terms of cylindrical components, it is often necessary to perform the **inverse transformation** — that is, to convert a vector expressed in cylindrical coordinates into its **rectangular (Cartesian)** form. This is especially useful when different quantities in an engineering problem are represented in different coordinate systems and need to be combined or compared in a common frame of reference.

Cylindrical $\rightarrow$ Rectangular Coordinate Transformation

To convert a vector given in **cylindrical components** into its **rectangular (Cartesian)** components, use the geometric relations between the two bases.

Unit-vector relations

$$\hat{\mathbf{a}}_x = \cos \phi \hat{\mathbf{a}}_\rho - \sin \phi \hat{\mathbf{a}}_\phi$$

$$\hat{\mathbf{a}}_y = \sin \phi \hat{\mathbf{a}}_\rho + \cos \phi \hat{\mathbf{a}}_\phi$$

$$\hat{\mathbf{a}}_z = \hat{\mathbf{a}}_z$$

Vector expressed in cylindrical form

$$\mathbf{A}(\rho, \phi, z) = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

Substituting the unit-vector relations gives the rectangular form:

$$\mathbf{A}(x, y, z) = (A_\rho \cos \phi - A_\phi \sin \phi) \hat{\mathbf{a}}_x + (A_\rho \sin \phi + A_\phi \cos \phi) \hat{\mathbf{a}}_y + A_z \hat{\mathbf{a}}_z$$

So the rectangular components are

$$A_x = A_\rho \cos \phi - A_\phi \sin \phi$$

$$A_y = A_\rho \sin \phi + A_\phi \cos \phi$$

$$A_z = A_z$$

The relationship between the **rectangular** and **cylindrical** components of a vector can be expressed as:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Recall that the **transformation matrix** $[T]$ used for the **Rectangular** $\rightarrow$ **Cylindrical** transformation is defined as:

$$[T] = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The inverse of this matrix, $[T]^{-1}$, is given by:

$$[T]^{-1} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence, the **Cylindrical** $\rightarrow$ **Rectangular** transformation can be compactly written as:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = [T]^{-1} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

or symbolically,

$$\mathbf{A}(x, y, z) = [T]^{-1} \mathbf{A}(\rho, \phi, z) \blacksquare$$

where:

$$[T] = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Physical Significance

This transformation is commonly encountered in **electromagnetics**, **fluid mechanics**, and **mechanical systems** where quantities such as the electric field, velocity, or stress are naturally defined in cylindrical coordinates but must be expressed in rectangular form for analysis, plotting, or computational purposes. For instance, in modeling the magnetic field around a current-carrying wire or the velocity field inside a cylindrical pipe, the field vectors are first described in cylindrical form but are often converted to rectangular coordinates for simulation or visualization.

Example: Converting Cylindrical Vector Components to Rectangular Components

Suppose a vector field represents the **magnetic flux density** at a certain point in space, given in cylindrical coordinates as

$$\mathbf{B}(\rho, \phi, z) = 2 \hat{\mathbf{a}}_\rho - 3 \hat{\mathbf{a}}_\phi + 4 \hat{\mathbf{a}}_z \text{ (in mT)}$$

Evaluate the components of this vector in the **rectangular coordinate system** at

$$\phi = 45^\circ.$$

Solution

Step 1. Recall the transformation relationships

From the cylindrical-to-rectangular component equations:

$$B_x = B_\rho \cos \phi - B_\phi \sin \phi$$

$$B_y = B_\rho \sin \phi + B_\phi \cos \phi$$

$$B_z = B_z$$

Step 2. Substitute the known values

Given:

$$B_r = 2, \quad B_\phi = -3, \quad B_z = 4, \quad \phi = 45^\circ$$

Compute the trigonometric values:

$$\cos 45^\circ = \sin 45^\circ = 0.7071$$

Step 3. Compute the rectangular components

$$B_x = (2)(0.7071) - (-3)(0.7071) = (1.4142) + (2.1213) = 3.5355$$

$$B_y = (2)(0.7071) + (-3)(0.7071) = (1.4142) - (2.1213) = -0.7071$$

$$B_z = 4$$

Step 4. Express the result in vector form

$$\mathbf{B} = 3.54 \hat{\mathbf{a}}_x - 0.71 \hat{\mathbf{a}}_y + 4.00 \hat{\mathbf{a}}_z \text{ (mT)} \blacksquare$$

Step 5. Matrix Representation

Alternatively, this conversion can be expressed compactly in **matrix form**:

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix}$$

At $\phi = 45^\circ$:

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3.5355 \\ -0.7071 \\ 4 \end{bmatrix}$$

which agrees with the direct computation.

Example

Suppose an electric field vector $\mathbf{E}$ is given by

$$\mathbf{E}(x, y, z) = 2\hat{\mathbf{a}}_x + x^2\hat{\mathbf{a}}_y + xy\hat{\mathbf{a}}_z$$

What is the field at the point $(1, 1, 2)$ expressed in cylindrical coordinates?

(See Figure 21)

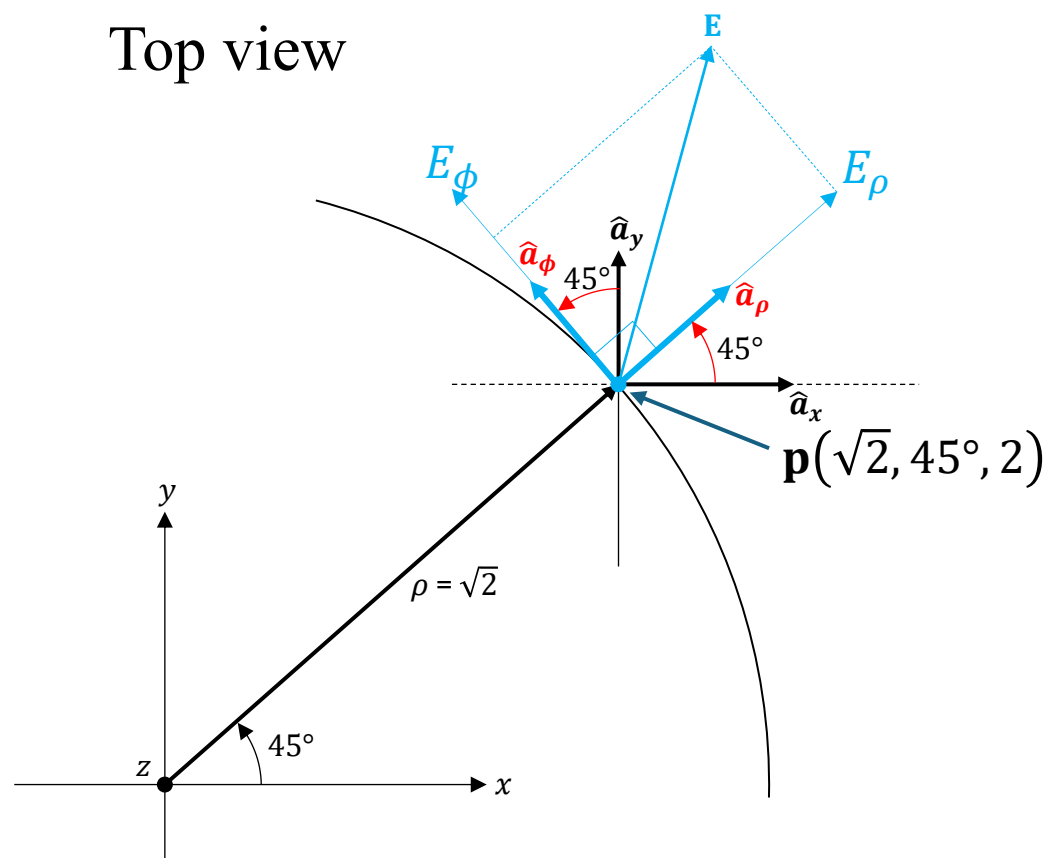


Figure 21

Solution

Step 1 — evaluate Cartesian components at the point

$$E_x(1, 1, 2) = 2, \quad E_y(1, 1, 2) = 1^2 = 1, \quad E_z(1, 1, 2) = 1 \cdot 1 = 1$$

So in Cartesian unit vectors:

$$\mathbf{E}(1, 1, 2) = 2\hat{\mathbf{a}}_x + \hat{\mathbf{a}}_y + \hat{\mathbf{a}}_z$$

Step 2 — find cylindrical coordinates of the point

$$\rho = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \phi = \tan^{-1} \left(\frac{1}{1} \right) = 45^\circ = \frac{\pi}{4}, \quad z = 2$$

Hence,

$$\mathbf{E}(\rho, \phi, z) = \mathbf{E}(\sqrt{2}, 45^\circ, 2)$$

Step 3 — compute rectangular components

$$\begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix} = \begin{bmatrix} \cos(45^\circ) & \sin(45^\circ) & 0 \\ -\sin(45^\circ) & \cos(45^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.1213 \\ -0.7071 \\ 1 \end{bmatrix}$$

Step 4 — final cylindrical expression

$$\mathbf{E} \left(\sqrt{2}, \frac{\pi}{4}, 1 \right) = 2.1213 \hat{\mathbf{a}}_\rho - 0.7071 \hat{\mathbf{a}}_\phi + \hat{\mathbf{a}}_z \blacksquare$$

Example: Magnetic Field in Cylindrical Coordinates

Problem

A long, straight wire carries a steady current $I = 5$ A in the $+z$ -direction (see Figure 22). From Ampère's law, the magnetic field around the wire in **rectangular coordinates** at a point (x, y, z) is given by

$$\mathbf{B}(x, y, z) = \frac{\mu_0 I}{2\pi(x^2 + y^2)} (-y \hat{\mathbf{a}}_x + x \hat{\mathbf{a}}_y),$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m.

Express $\mathbf{B}$ in **cylindrical coordinates** at the point $(\rho, \phi, z) = (0.1 \text{ m}, 45^\circ, 0 \text{ m})$.

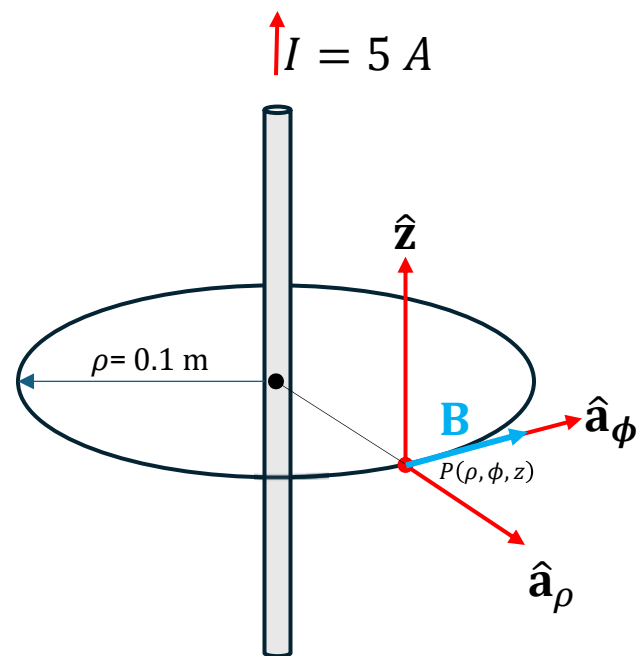


Figure 22

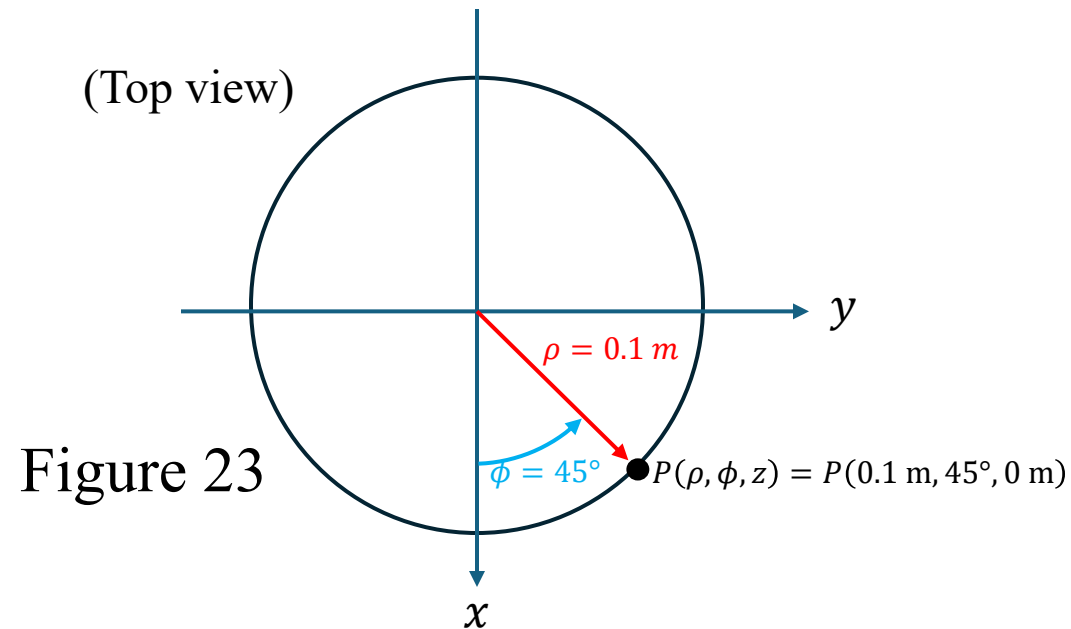
Solution

1. Convert the point to Cartesian

$$x = \rho \cos \phi = 0.1 \cos 45^\circ = 0.1 \times \frac{\sqrt{2}}{2} = 0.0707107 \text{ m},$$

$$y = \rho \sin \phi = 0.0707107 \text{ m}.$$

(see Figure 23)



2. Evaluate the scalar prefactor

$$\frac{\mu_0 I}{2\pi(x^2 + y^2)} = \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.01} = \frac{2 \times 10^{-6}}{0.01} = 2.0 \times 10^{-4} \text{ (SI).}$$

3. Compute Cartesian components

$$\begin{aligned}\mathbf{B} &= 2.0 \times 10^{-4}(-y \hat{\mathbf{a}}_x + x \hat{\mathbf{a}}_y) \\ &= 2.0 \times 10^{-4}(-0.0707107 \hat{\mathbf{a}}_x + 0.0707107 \hat{\mathbf{a}}_y) \\ \Rightarrow \mathbf{B} &\approx -1.4142 \times 10^{-5} \hat{\mathbf{a}}_x + 1.4142 \times 10^{-5} \hat{\mathbf{a}}_y \text{ T.}\end{aligned}$$

Step 4 — Compute cylindrical components

The cylindrical components (B_ρ, B_ϕ, B_z) are related to the rectangular components (B_x, B_y, B_z) by

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix}$$

At $\phi = 45^\circ$, $\cos 45^\circ = \sin 45^\circ = 0.707$ we obtain

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} 0.7071 & 0.7071 & 0 \\ -0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1.4142 \times 10^{-5} \\ 1.4142 \times 10^{-5} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2.0 \times 10^{-5} \\ 0 \end{bmatrix}$$

Therefore, the cylindrical components are

$$B_\rho = 0 \text{ T}$$

$$B_\phi = 2.0 \times 10^{-5} \text{ T}$$

$$B_z = 0 \text{ T}$$

Final answer:

$$\mathbf{B} = 2.0 \times 10^{-5} \hat{\boldsymbol{\phi}} \text{ T} \blacksquare$$

Example

An electric field vector in cylindrical coordinates at the point $(x, y, z) = (1, 3, 5)$ is given by

$$\mathbf{E}(\rho, \phi, z) = 0.17 \hat{\mathbf{a}}_\rho - 0.508 \hat{\mathbf{a}}_\phi - 2.54 \hat{\mathbf{a}}_z \text{ V/m}$$

Determine the electric field expressed in rectangular (Cartesian) coordinates.
(see Figure 24)

Top view

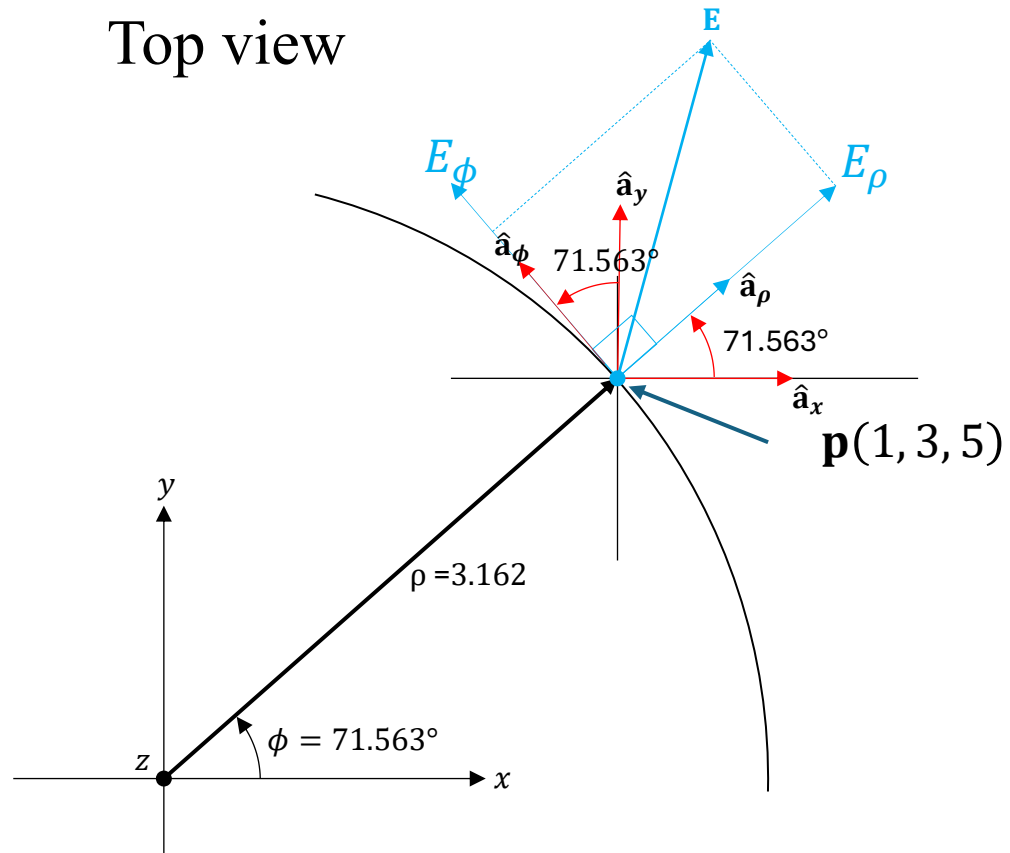


Figure 24

Solution

Step 1. Find the cylindrical angle ϕ

$$\tan \phi = \frac{y}{x} = \frac{3}{1} = 3$$

$$\Rightarrow \phi = \tan^{-1}(3) = 71.565^\circ$$

(see Figure 25)

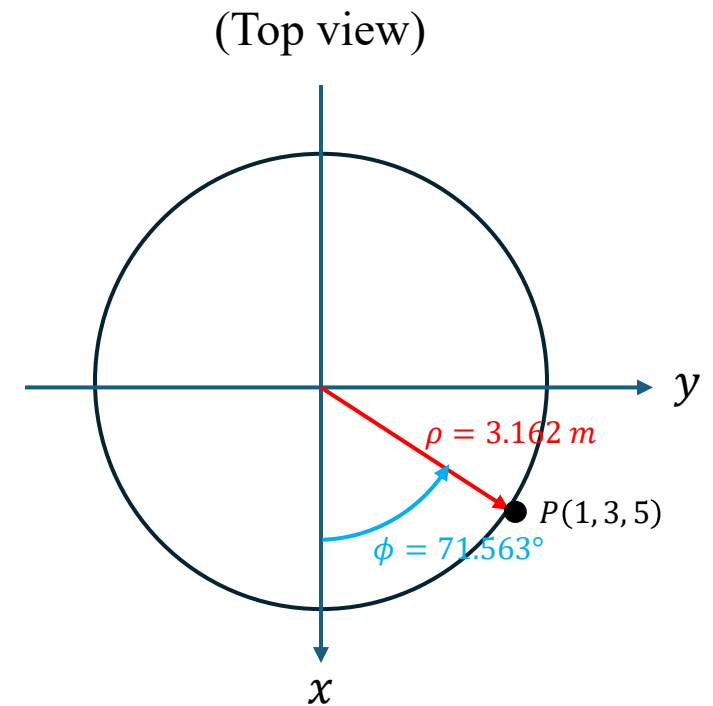


Figure 25

Step 2. Express cylindrical unit vectors in terms of rectangular unit vectors

$$\hat{\mathbf{a}}_\rho = \cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_\phi = -\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_z = \hat{\mathbf{a}}_z$$

Step 3. Substitute into the field expression

$$\mathbf{E} = 0.17(\cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y) - 0.508(-\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y) - 2.54 \mathbf{a}_z$$

In matrix form,

$$\begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix}$$

Step 4. Substitute numerical values

$$\cos(71.565^\circ) = 0.3162,$$

$$\begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix} = \begin{bmatrix} 0.17 \\ -508 \\ -2.54 \end{bmatrix} \sin(71.565^\circ) = 0.9487$$

$$\begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = \begin{bmatrix} 0.3162 & -0.9487 & 0 \\ 0.9487 & 0.3162 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.17 \\ -508 \\ -2.54 \end{bmatrix}$$

Summary

- Cylindrical coordinates (ρ, ϕ, z) are suited for systems with **cylindrical symmetry**.
- They link directly with rectangular coordinates through simple relations.
- Coordinate surfaces, unit vectors, and differential elements make them powerful tools for solving engineering problems.
- Widely applied in electromagnetics, fluid mechanics, and heat transfer.

END OF UNIT 3